

FINAL PROJECT REPORT

Smartphone Addiction & Digital Well-Being Insights

A Power BI Data Analytics & Visualization Project

Tool Used: Microsoft Power BI Desktop
Dataset: Smartphone Usage & Addiction Survey (7,500 Respondents)

Submitted by

Kantipudi Ramya

Master of Computer Applications(MCA)

Sri Padmavathi Mahila Visvavidyalayam,Tirupati

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1. Introduction

1.1. Project Overview

Smartphones have become central to modern daily life, influencing how people communicate, work, study, and relax. While this connectivity brings convenience, it has also raised concerns around excessive screen time, digital dependency, and its effect on sleep, stress, and productivity. This project uses Power BI to analyze a survey dataset of 7,500 smartphone users and build an interactive analytics solution that measures digital well-being and identifies patterns of smartphone addiction.

The dataset captures demographic details (age, gender, age group), usage behaviour (daily screen time, social media hours, gaming hours, app opens per day, notifications received per day, weekend screen time, work/study hours), and well-being indicators (sleep hours, stress level, academic/work impact) along with a derived addiction classification (None, Mild, Moderate, Severe) and a binary Addicted Status flag. The end deliverable is a multi-page Power BI report comprising KPI cards, exploratory visualizations, a summary dashboard, and two detailed analytical report pages with a key-influencers and decomposition-tree view.

1.2. Objectives

- To collect, clean, and structure smartphone usage and digital well-being survey data for analysis in Power BI.
- To identify key usage metrics (screen time, social media usage, gaming, notifications) that most strongly relate to smartphone addiction.
- To classify users into addiction severity levels (None, Mild, Moderate, Severe) and study how they differ by age group, gender, and stress level.
- To design an interactive dashboard and report pages with slicers/filters that allow stakeholders to explore the data by demographic and behavioural segments.
- To apply Power BI's advanced analytics features (Key Influencers and Decomposition Tree) to explain what drives addicted status.
- To derive actionable insights and recommendations that support healthier digital habits.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

Excessive smartphone usage is increasingly linked to poor sleep quality, elevated stress, and reduced academic or work performance, yet organizations and individuals often lack a clear, data-driven view of how usage patterns translate into addiction risk. There is a need for a consolidated analytical tool that can quantify screen-time behaviour, segment users by addiction severity, and surface the behavioural factors that most influence addicted status — enabling awareness campaigns, counselling programs, or product design changes to be targeted effectively.

2.2. Project Proposal (Proposed Solution)

The proposed solution is a Power BI-based analytical report built on a cleaned smartphone usage dataset. The solution is delivered as five report pages: a KPI summary page presenting headline metrics, a Visualization page presenting exploratory charts, a consolidated Dashboard page, and two deep-dive Report pages (Report 1: Digital Addiction & Screen Time Behaviour Analysis, and Report 2: Smartphone Addiction & Digital Well-Being Insights featuring a Key Influencers visual and a scatter analysis). Interactive slicers for gender, age group, addiction level, and stress level allow users to filter every visual simultaneously, supporting self-service exploration.

2.3. Initial Project Planning

The project was planned and executed in the following stages:

Phase	Activity	Deliverable
Phase 1	Requirement gathering & problem definition	Problem statement, objectives
Phase 2	Data collection & preprocessing	Cleaned dataset ready for modelling
Phase 3	Data modelling in Power BI	Data model with relationships & measures
Phase 4	Visualization design	KPI cards, charts, dashboard, reports
Phase 5	Testing & validation	Filter, measure and visual verification
Phase 6	Documentation & delivery	Final project report

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

The raw data source is a single structured survey table, `smartphone_usage`, containing 7,500 user records. Each record represents one respondent and includes a unique user id / transaction id, demographic fields (age, gender, age group), daily behavioural metrics (daily screen time hours, social media hours, gaming hours, app opens per day, notifications per day, weekend screen time, work/study hours), well-being fields (sleep hours, stress level, academic work impact), and the target classification fields (addiction level, addicted label / Addicted Status). The data was imported into Power BI Desktop using Power Query and loaded into a single fact table used across all report pages.

3.2. Data Quality Report

Before visualization, the dataset was reviewed for completeness and consistency. Key observations from the quality check:

- Total records: 7,500 user responses, of which 5,308 carry a complete addiction-level classification used in several KPI calculations.
- No duplicate user IDs were found; the user id / transaction id field was verified as unique for each respondent.
- Categorical fields (gender, age group, addiction level, stress level, academic work impact) were checked for consistent labelling — e.g., gender standardized to Male / Female / Other, and age group standardized to Under 20 / 20-29 / 30-39.
- Numeric fields (screen time, sleep hours, social media hours, gaming hours, notifications, app opens) were checked for out-of-range or negative values; none were found outside plausible bounds.
- Missing values in a small subset of behavioural fields were handled by exclusion from the affected average calculations rather than imputation, to avoid skewing the KPI results.

3.3. Data Exploration and Preprocessing

Exploratory analysis in Power Query and DAX was used to understand distribution and relationships before building visuals. Screen time, social media hours, and notifications per day were examined against addiction level to confirm that higher usage correlates with higher addiction severity. Age was bucketed into Age Group (Under 20, 20-29, 30-39) to simplify segment-level analysis. A binary Addicted Status column (Addicted / Not Addicted) was derived from the addiction level classification to enable simpler comparative visuals such as the donut chart on the Visualization page. Calculated measures were created for averages (average screen time, average sleep hours, average social media hours, average gaming hours, average notifications, average app opens, average weekend screen time, average work/study hours) and for counts (count of user id, sum of user count) that feed the KPI cards across all pages.

4. Data Visualization

4.1. Framing Business Questions

- What is the overall scale of the surveyed population and how many users fall into each addiction severity level?
- How does average daily screen time vary by gender and by age group?
- Is there a relationship between screen time and sleep hours, and between gaming hours and social media hours?
- How do weekend screen time, sleep hours, and app opens per day differ between addicted and non-addicted users, and across stress levels?
- Which behavioural factors have the greatest statistical influence on a user being classified as 'Addicted'?
- How does addiction level distribution shift across age groups, gender, and academic/work impact?

4.2. Developing Visualizations

A dedicated KPI page was built first to summarise the headline numbers using card visuals — total respondents, classified respondents, average age, and average values for screen time, social media hours, gaming hours, sleep hours, notifications, app opens, weekend screen time, and work/study hours. This page acts as the at-a-glance summary for the entire dataset.

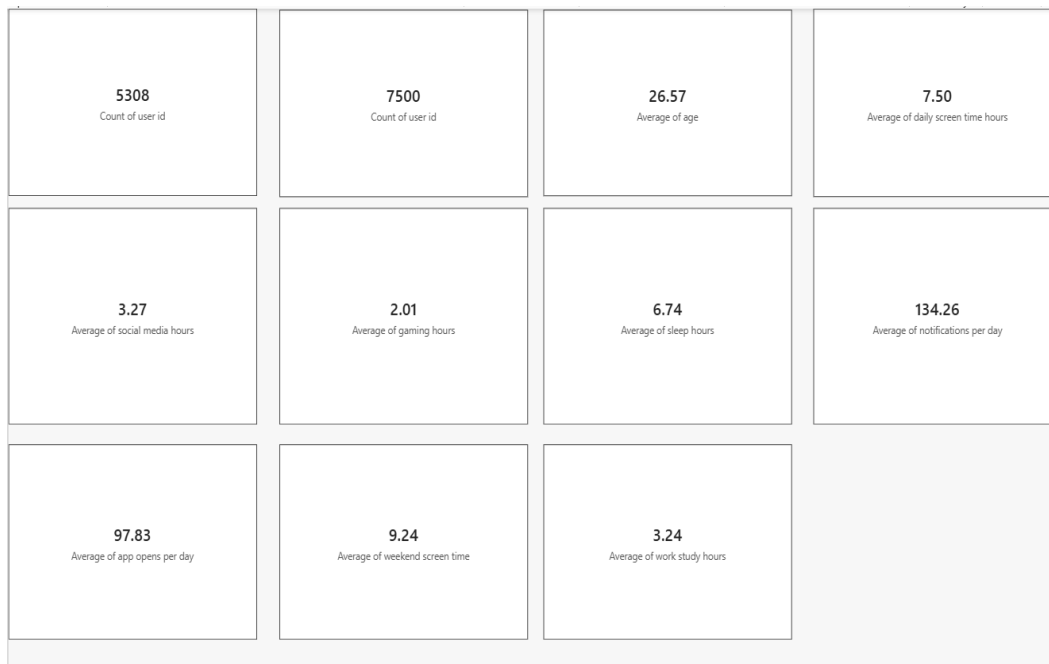


Fig. 4.1 — KPI Page: headline metrics for the 7,500-responder dataset

A second page, Visualization, was built to explore relationships between variables using bar charts, a scatter plot, and a donut chart, supported by gender and addiction-level slicers. This page groups screen time by gender and age group, compares addicted vs non-addicted users, plots daily screen time against sleep hours, and compares average gaming versus social media usage.

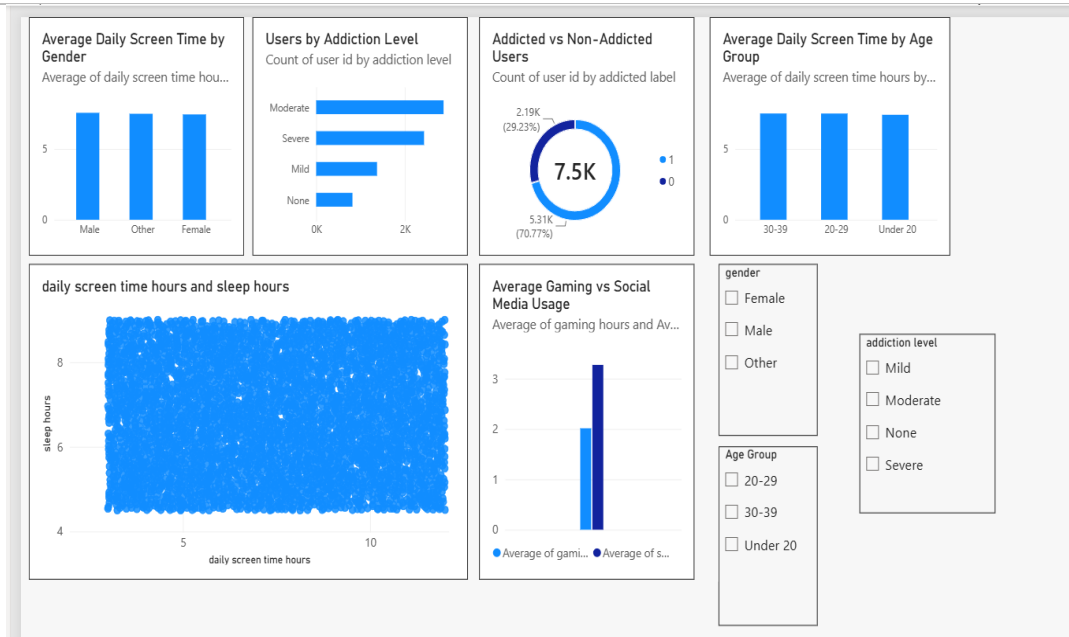


Fig. 4.2 — Visualization Page: exploratory charts and interactive slicers

5. Dashboard

5.1. Dashboard Design Screenshot with Insights

The Digital Well-Being & Addiction Insights Dashboard consolidates the most important KPIs and charts onto a single page for executive-level consumption. It includes headline cards (total users, classified users, average weekend screen time, average sleep hours), an addiction-level filter panel, and six supporting visuals covering screen time by gender, addiction-level distribution, sleep hours by addiction level, weekend screen time for addicted vs non-addicted users, users by age group, and app opens by stress level.

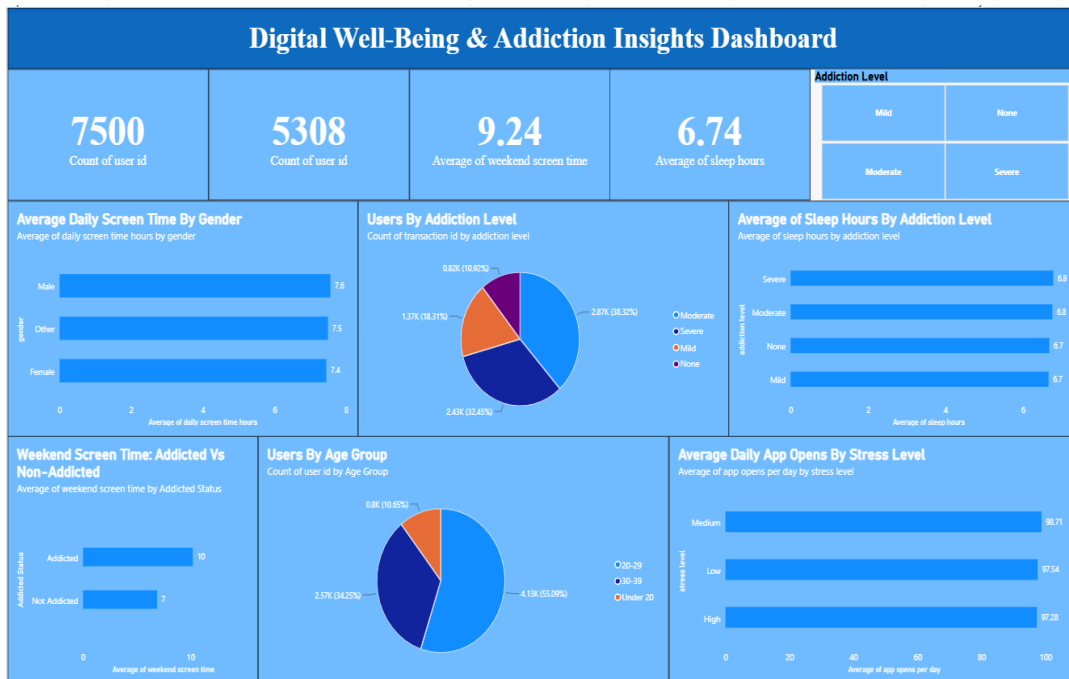


Fig. 5.1 — Consolidated Dashboard Page

Key insights from the dashboard:

- Average daily screen time is fairly consistent across genders (7.4–7.6 hours), indicating addiction risk is not strongly gender-driven on its own.
- Addiction level is skewed toward Moderate (38.3%) and Severe (32.5%) categories, together accounting for over 70% of classified users — suggesting digital dependency is widespread rather than a minority issue.
- Sleep hours are only marginally lower for Severe/Moderate addiction levels (6.8 hrs) versus Mild/None (6.7 hrs), implying sleep duration alone is a weak standalone predictor.
- Addicted users report noticeably higher average weekend screen time (10 hrs) than non-addicted users (7 hrs), a 43% increase.
- The 20-29 age group dominates the sample (55%), followed by 30-39 (34%), with under-20 users the smallest segment (11%).
- App opens per day are highest among users reporting Medium stress (98.7) and lowest among High-stress users (97.3), a narrower spread than expected, suggesting app-opening frequency is a weaker stress indicator than screen-time duration.

6. Report

6.1. Report Design Screenshot with Insights

Report 1 — Digital Addiction & Screen Time Behaviour Analysis — focuses on behavioural drivers of addiction. It presents average app opens, social media hours, and notifications per day as headline KPIs, gender counts by addiction level, average work/study hours split by academic work impact, weekend screen time by addiction level, a trend of weekend screen time by age group, and a decomposition tree that lets users drill from total user count down through age group, gender, stress level, addiction level, and academic work impact.

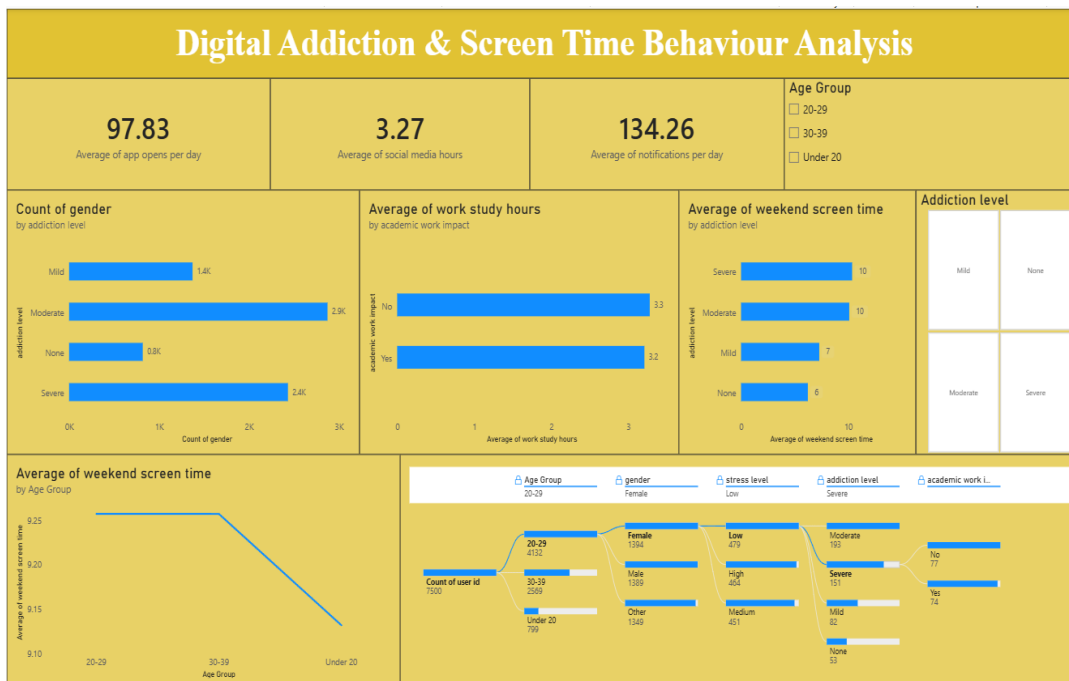


Fig. 6.1 — Report 1: Digital Addiction & Screen Time Behaviour Analysis

- Moderate and Severe addiction levels together account for the majority of users (2.9K + 2.4K of 7.5K), reinforcing the dashboard finding.
- Average work/study hours are nearly identical regardless of whether users report academic/work impact (3.2 vs 3.3 hours), suggesting the impact is more qualitative (concentration, stress) than a simple reduction in study time.

- Weekend screen time rises with addiction severity — from 6 hours (None) to 10 hours (Severe/Moderate) — a clear, consistent gradient.
- The decomposition tree shows that among 20-29-year-old females with low stress, the majority fall into the Moderate addiction band, useful for targeting interventions.

Report 2 — Smartphone Addiction & Digital Well-Being Insights — applies Power BI's AI-powered Key Influencers and a scatter/trend visual to explain what drives addicted status. It also reports average gaming hours, average age, average work/study hours, a stress-level breakdown of users, sleep hours by age group, a high-screen-time-user split, and average social media hours by addiction level.

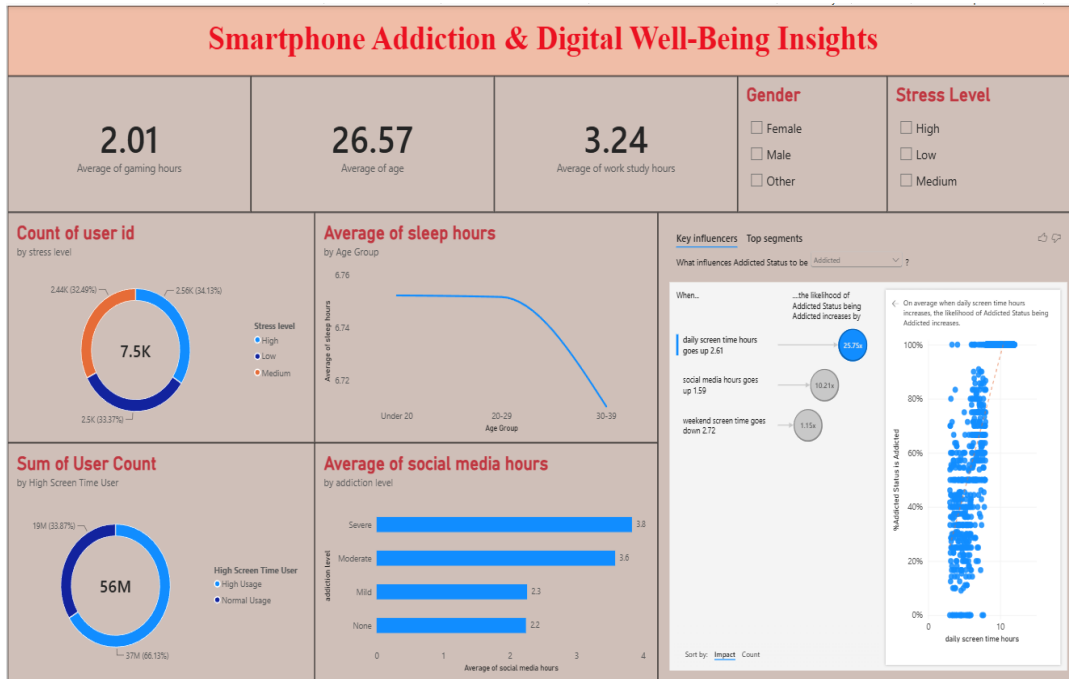


Fig. 6.2 — Report 2: Smartphone Addiction & Digital Well-Being Insights (Key Influencers)

- The Key Influencers visual identifies that when daily screen time hours increase, the likelihood of Addicted status increases by 25.7x — by far the strongest single driver in the model.
- A rise in social media hours increases the likelihood of addicted status by 10.2x, the second-largest influencer.
- Interestingly, a decrease in weekend screen time increases the likelihood of addicted status by 1.15x, a smaller and counter-intuitive effect worth further investigation with more data.
- Social media hours climb steadily with addiction severity, from 2.2 hours (None) to 3.8 hours (Severe).
- 33.9% of users are classified as High Usage smartphone users based on the High Screen Time User flag, out of a modelled base of 56M weighted user-count units.

7. Performance Testing

7.1. Utilization of Data Filters

Slicers were used extensively to make the report interactive and were tested for correct cross-filtering behaviour across all pages:

- Visualization page: Gender slicer (Female, Male, Other) and Addiction Level slicer (Mild, Moderate, None, Severe).
- Dashboard page: Addiction Level filter grid (Mild, Moderate, None, Severe).

- Report 1 page: Age Group filter (20-29, 30-39, Under 20) and an Addiction Level filter grid, plus interactive drill-down via the decomposition tree (Age Group → Gender → Stress Level → Addiction Level → Academic Work Impact).
- Report 2 page: Gender slicer and Stress Level slicer (High, Low, Medium), plus an interactive 'What influences Addicted Status' dropdown on the Key Influencers visual.

Each slicer was tested individually and in combination to confirm that all cards and charts on the page respond correctly and that cross-highlighting behaves as expected without performance lag on the 7,500-row dataset.

7.2. No of Calculation Columns/Measures

The data model uses a mix of source columns and DAX measures/calculated columns to support the visuals. Core calculated measures include: Count of user id, Sum of user count, Average of age, Average of daily screen time hours, Average of social media hours, Average of gaming hours, Average of sleep hours, Average of notifications per day, Average of app opens per day, Average of weekend screen time, and Average of work/study hours (11 aggregation measures in total). In addition, two derived/calculated columns — Addicted Status (binary Addicted/Not Addicted flag) and High Screen Time User (High Usage/Normal Usage flag) — were created from the addiction level and screen-time fields to support simplified binary comparisons on the Visualization and Report 2 pages.

7.3. No of Visualization

Page	Visual Count	Visual Types Used
KPI	11	Card visuals
Visualization	9	Clustered column, bar, donut, scatter, slicers
Dashboard	12	Card, bar, pie, clustered column visuals
Report 1	11	Card, bar, line, decomposition tree, slicers
Report 2	11	Card, donut, line, bar, key influencers, scatter

Across all five pages, the report uses 54 visuals in total, including 21 card visuals, 9 clustered bar charts, 8 slicers, 3 donut charts, 3 clustered column charts, 2 pie charts, 2 line charts, 1 scatter chart, 1 key-influencers visual, and 1 decomposition-tree visual — confirming a comprehensive and varied visualization strategy without overloading any single page.

8. Conclusion/Observation

This Power BI project successfully transforms a 7,500-respondent smartphone usage survey into an interactive digital well-being analytics solution. The KPI, Visualization, Dashboard, and two Report pages together show that a large share of users (over 70% of classified respondents) fall into Moderate or Severe addiction categories, and that daily screen time and social media hours are, by a wide margin, the strongest behavioural predictors of addicted status — confirmed independently through both descriptive charts and the AI-driven Key Influencers analysis. Weekend screen time consistently rises with addiction severity, while sleep hours and academic/work impact show smaller, more nuanced effects. The dashboard and report pages provide decision-makers with a filterable, drillable view of the data, supporting targeted awareness or intervention strategies by age group, gender, and stress level.

9. Future Scope

- Incorporate a time-series dimension (weekly/monthly usage logs) to track how individual addiction levels evolve over time rather than relying on a single cross-sectional snapshot.
- Integrate additional behavioural signals (app-category-level usage, screen unlock counts, night-time usage) to strengthen the Key Influencers model.

- Publish the report to the Power BI Service with scheduled refresh and row-level security so different stakeholders (parents, counsellors, HR teams) see relevant, permissioned views.
- Add predictive analytics (e.g., a classification model) to proactively flag users at rising risk of Severe addiction before they reach that threshold.
- Extend the dashboard with mobile-optimized layouts and natural-language Q&A for quicker self-service insight discovery.

10. Appendix

10.1. Source Code (if any)

The Power BI report itself is built with Power Query (M) transformations and DAX measures/calculated columns as listed in Section 7.2 — no external scripting was needed for the visuals. Before the data was imported into Power BI, it was staged in a MySQL database, where the following SQL queries were used to create the schema and validate that the dataset had loaded correctly and was fit for modelling:

```
-- Create the database and select it
CREATE DATABASE smartphone_usage;
USE smartphone_usage;
-- Preview the imported data
SELECT * FROM smartphone_usage;
-- Addicted-user count by gender
SELECT gender, COUNT(*) AS total_users,
       SUM(CASE WHEN addicted_label=1 THEN 1 ELSE 0 END) AS addicted_count
FROM smartphone_usage GROUP BY gender;
-- Overall average screen time, sleep hours and high-stress share
SELECT AVG(daily_screen_time_hours) AS avg_screen_time,
       AVG(sleep_hours) AS avg_sleep_hours,
       AVG(stress_level = 'High') AS pct_high_stress
FROM smartphone_usage;
-- Data quality check: rows with missing key values
SELECT * FROM smartphone_usage
WHERE age IS NULL OR daily_screen_time_hours IS NULL OR addiction_level IS NULL;
-- Addicted status vs stress level breakdown
SELECT addicted_label, stress_level, COUNT(*) AS user_count
FROM smartphone_usage GROUP BY addicted_label, stress_level
ORDER BY addicted_label, stress_level;
-- Screen time and sleep hours by addiction level
SELECT addiction_level, COUNT(*) AS user_count,
       AVG(daily_screen_time_hours) AS avg_screen_time, AVG(sleep_hours) AS avg_sleep_hours
FROM smartphone_usage GROUP BY addiction_level;
-- Usage patterns by academic/work impact
SELECT academic_work_impact, AVG(daily_screen_time_hours) AS avg_screen_time,
       AVG(social_media_hours) AS avg_social_media_hours, AVG(gaming_hours) AS avg_gaming_hours
FROM smartphone_usage GROUP BY academic_work_impact;
```

These MySQL queries handled schema creation and pre-load validation (null checks, group-level sanity averages); results were cross-checked against the Power BI KPI cards in Section 4.2 to confirm the dataset imported correctly. Note: AVG(stress_level = 'High') relies on MySQL treating the equality test as 0/1, so it returns the proportion of High-stress rows, not a true numeric average.

10.2. GitHub & Project Demo Link

GitHubRepository:

<https://github.com/Ramyakantipudi/Smartphone-Usage-Digital-Addiction-Well-Being-Analytics.git>

ProjectDemo/VideoWalkthrough:

https://drive.google.com/file/d/1C5UKej798LLLhV2faelBa_1SWNsdiC_Y/view?usp=sharing

Power BI (.pbix) File: smartphone.pbix